Sleep, Study Habits, Stress, and Academic Performance: Analyzing the Impact of Student Lifestyle Habits on GPA

MATH 189 Project Proposal

Motivation and Problem Statement

College students at UCSD and across the country often sacrifice sleep in order to keep up with academic demands, especially during midterm and final examination seasons. Many students believe that studying longer hours and staying up late will improve their performance; however, this leads to poor sleeping habits whose negative impact can affect memory, concentration, and stress levels, making overall productivity less efficient.

As academic pressures continue to increase, it is important to better understand how sleep, study habits, and stress contribute to academic success. This topic is especially relevant because mental health concerns, burnout, and sleep deprivation are becoming increasingly common among college students.

By analyzing the relationships between sleep patterns, study behaviors, stress levels, and academic performance, this project aims to provide insight into whether sacrificing sleep truly benefits students academically, or whether healthier habits lead to better outcomes. The findings could help students make more informed decisions about time management, studying strategies, and priority-setting during demanding academic periods.

Data Sources and Acquisition

The dataset for the project will come from publicly available datasets related to student sleep habits, stress levels, study behaviors, and academic performance. Specifically, we plan to use the *Student Stress Factors* dataset and the *Sleep Health and Lifestyle* dataset available on Kaggle, supplemented by openly released university-based research datasets on student sleep and GPA. These datasets include variables such as sleep duration, sleep quality, screen time, stress levels, study hours, productivity outcome, and GPA.

Using publicly available datasets allows us to work with larger and more diverse samples, and it will make the project more feasible within the time frame of the course. Moreover, these datasets provide enough variables and observations to perform meaningful analysis and observe trends between student behaviors and academic performance. Where variable definitions differ across sources, we will standardize them during preprocessing.

The data will be cleaned and organized before analysis to handle missing values and inconsistent entries, which will improve the accuracy and reliability of the results.

Proposed Analyses

Our analysis will proceed in three stages. First, we will use descriptive statistics and exploratory visualizations to summarize trends in sleep habits, stress levels, and academic outcomes across the cleaned dataset, and to identify any preliminary patterns or anomalies.

Second, we will conduct correlation analysis to quantify relationships between key variable pairs. For example, sleep duration and GPA, or self-reported stress levels, anxiety score, and sleep time. We will also apply hypothesis testing and group comparison tests (e.g., two-sample t-tests, Mann-Whitney U) to determine whether students with healthier sleep habits perform statistically differently from those who do not.

Third, we will fit a multiple linear regression model to jointly predict GPA as a function of sleep duration, stress levels, and study hours, allowing us to assess the unique contribution of each variable while controlling for the others. To test for the nonlinear effects hypothesized above, we will incorporate polynomial and interaction terms into the model. We will evaluate model fit using R^2 and residual diagnostics, and use the F-test and t-tests on individual coefficients to assess statistical significance. This multi-step modeling approach allows us to move beyond simple pairwise correlations to a more complete picture of how lifestyle factors jointly influence academic outcomes.

Expected Outcomes

We expect students with healthier sleep habits and lower stress levels will tend to perform better academically than students who experience chronic sleep deprivation and high stress. However, there might exist upper and lower limits to this relationship: once stress exceeds a certain level, or once it is too low, academic performance may actually decline or stagnate. It is also expected that there is a negative correlation between sleep duration and stress levels. We expect there is an optimal amount of sleep and an optimal stress level, which means balanced study habits and consistent sleep schedules may be associated with the best academic outcomes.

If the regression analyses confirm these patterns, the results will provide statistically grounded guidance for students on how to structure their routines. More broadly, these findings will contribute to the growing body of evidence linking student lifestyle habits to academic success, and will be presented as interactive visualizations and a concise written report accessible to a general student audience. Ultimately, we hope these results will encourage students to adopt healthier routines and improve both their academic and personal lives.